

## CODING & SOLUTION

|                      |               |
|----------------------|---------------|
| <b>Date</b>          | 28 June 2026  |
| <b>Team ID</b>       |               |
| <b>Project Name</b>  | Rising Waters |
| <b>Maximum Marks</b> | 5 Marks       |

### Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

### Solution Summary

| Field                         | Details                                                                                                                                                                                                                                                                                |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository Link / URL         | <a href="https://github.com/doprajnasai/Flood-Prediction-System">https://github.com/doprajnasai/Flood-Prediction-System</a>                                                                                                                                                            |
| Programming Language(s)       | Python, HTML, CSS                                                                                                                                                                                                                                                                      |
| Framework(s) Used             | Flask, Scikit-learn, XGBoost                                                                                                                                                                                                                                                           |
| Key Features Implemented      | Real-time flood prediction, XGBoost ML model, Flask web application, responsive UI, input validation, model deployment using Joblib                                                                                                                                                    |
| Pending / Incomplete Features | No everything mentioned are implemented in this project.                                                                                                                                                                                                                               |
| Setup / Run Instructions      | Clone repository → Install requirements.txt → Run python app.py → Open <a href="http://127.0.0.1:5000">http://127.0.0.1:5000</a> or use the deployed Render link ( <a href="https://flood-prediction-system-3fv8.onrender.com">https://flood-prediction-system-3fv8.onrender.com</a> ) |

## Code Quality Checklist

| S.No | Criteria                                               | Status (Yes / No) |
|------|--------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions / classes | Yes               |
| 2    | Meaningful variable and function names are used        | Yes               |
| 3    | Code includes comments / documentation where necessary | Yes               |
| 4    | Error handling is implemented for critical operations  | Yes               |
| 5    | The application runs without critical errors           | Yes               |
| 6    | Code is committed to a version control repository      | Yes               |

### Additional Notes / Comments:

The project follows a complete machine learning workflow from data preprocessing to deployment. The application is deployed using Render, uses an XGBoost model for prediction, and provides a responsive Flask-based web interface with input validation and real-time flood prediction.